Faizan Farhad

AI / ML Engineering Intern

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LinkedIn | GitHub | Portfolio

PROFILE

BCA student focused on AI/ML and AI engineering, with hands-on experience building LLM-powered agents, NLP applications, and AI-driven data tools. Experienced in developing natural-language-to-SQL systems, browser automation workflows, and data analysis applications. Interested in building reliable AI systems that solve practical problems through automation, intelligent reasoning, and tool integration.

EDUCATION

Bachelor of Computer Applications (BCA)
CPJ College 2024 – 2028(Expected) | Narela, Delhi

SKILLS

- **Languages**: Python, SQL
- **AI / ML & Agents:** PyTorch, Hugging Face Transformers, BitsAndBytes, Scikit-Learn, Ollama, Prompt Engineering, Agentic Workflows
- **Backend & Web:** FastAPI, Uvicorn, REST APIs, HTTPX, SQLAlchemy, OAuth 2.0
- **NLP:** Text Preprocessing, Tokenization
- **Deep Learning**: PyTorch, CNN, Neural Networks
- **Databases:** SQLite

LANGUAGES

- Hindi
- English

PROJECTS

Browser Automation Agent [Sep 2026 – Oct 2026]

• Engineered an LLM-driven browser automation agent that converts natural-language tasks into structured action plans and executes them through Playwright.
• Designed a planner–validator–executor architecture with bounded replanning, validating generated actions against live browser state before execution and using DOM observations to recover from invalid plans.
• Engineered an autonomous LLM agent executing structured web tasks through an API-first router, executing direct HTTP requests for read operations and Playwright for complex DOM interactions.
• Integrated Chrome DevTools Protocol (CDP) for persistent session attachment and desktop Google OAuth 2.0 to automate authenticated user workflows (e.g., Gmail).
• Implemented runtime state reliability via atomic JSON snapshots (os.replace), logging step-by-step attempt histories, failed DOM observations, and execution traces
Tech Stack: Python, Playwright, CDP (Chrome DevTools Protocol), Ollama, DeepSeek API, HTTPX, OAuth 2.0

QueryCraft — End-to-End Plain English to SQL Engine [Aug 2026 – Sep 2026]

• Built and deployed a full-stack NL-to-SQL FastAPI engine enabling natural language querying over relational databases with automated schema introspection via SQLAlchemy
• Engineered a multi-tier safety pipeline enforcing SQLite PRAGMA query_only=ON for read-only isolation, blocking destructive syntax (GRANT, EXEC), and stripping model reasoning fences.
• Implemented a dual-generation engine supporting 4-bit quantized local models (BitsAndBytes, beam search) and cloud APIs (DeepSeek, OpenAI), delivering **sub-second inference (<1s)** via API and caching local model weights
• Achieved **89–90% query execution accuracy** validated against a deterministic 12-case automated evaluation suite testing aggregation, multi-table joins, and ambiguous prompts.
Tech Stack: FastAPI, Python, SQLAlchemy, SQLite, Hugging Face Transformers, BitsAndBytes, PyTorch, DeepSeek/OpenAI API

COURSES

AI/ML | Daksh Gurukul (IIT Guwahati)

June 2025 – Oct 2026

• Acquired deep theoretical foundations and practical applications in AI/ML principles, deep learning frameworks, and predictive modeling techniques to solve real-world problems.